Math 107 Final exam practice

Last updated: October 8, 2026.

Cumulative practice for the Final exam · Wednesday, December 16, 10:20 a.m.–12:10 p.m., FR 201. Use with the study guide. Also useful: Midterm 1 practice, Midterm 2 practice.

Four blocks: **(A)** Unit I refresh, **(B)** Unit II refresh, **(C)** since Midterm 2, **(D)** mixed cumulative. Answers at the end.

A. Unit I refresh (Midterm 1 style)

Vectors, geometry, linear functions, neighbors / clustering, span and basis. Show work on computations.

Definitions and meaning

A1. What is a **linear combination** of vectors $a_1, \dots, a_k$?

A2. Define the **inner product** of two n-vectors. When are two vectors **orthogonal**?

A3. When is $f\colon \mathbb{R}^n \to \mathbb{R}$ **linear**? How does **affine** differ?

A4. Write formulas for $\|x\|$ and the distance between x and y. Write cosine similarity of nonzero a and b.

A5. What is a **basis** of a subspace V? What is $\dim V$?

A6. True or false, with a one-line reason: "If $\|a - b\|$ is small, then $\angle(a, b)$ must be near 0."

A7. For each expression, say whether the result is a **scalar**, an n**-vector**, a **set of vectors**, or **undefined** (brief reason if undefined). Assume sizes make sense unless noted.

(a) $a^\mathsf{T} b$ for n-vectors a, b

(b) $\operatorname{span}\{a, b\}$

(c) $a^\mathsf{T} b + c$ when c is an n-vector

(d) unit vector $a/\|a\|$ when $a \neq 0$

Computational

A8. Let $u = (3, -1, 2)$ and $v = (0, 4, -1)$. Compute $u + v$, $2u - v$, and $u^\mathsf{T} v$.

A9. Write $w = (5, 1)$ as a linear combination of $a = (1, 0)$ and $b = (1, 1)$.

A10. Let $x = (2, 1, -1)$. Compute $\|x\|$ exactly. Distance from $p = (1, 1)$ to $q = (4, -3)$?

A11. Cosine of the angle between $a = (1, 0)$ and $b = (1, 1)$. Unit vector in the direction of $(3, 4)$?

A12. Is $f(x) = 3x_1 - x_2 + 2$ linear or affine? Brief reason.

A13. Query $x = (1, 2)$; candidates $z_1 = (0, 2)$, $z_2 = (3, 2)$, $z_3 = (1, 3)$. Which is nearest in Euclidean distance? Give the three distances.

A14. Points $x_1 = (0, 0)$, $x_2 = (0, 2)$, $x_3 = (6, 0)$, $x_4 = (6, 2)$; representatives $z_1 = (1, 1)$, $z_2 = (5, 1)$. Assign each point to the nearest z_j, then update each z_j to the mean of its points.

A15. Are $(1, 2)$ and $(-2, -4)$ linearly independent? Give a dependence relation or argue independence.

A16. Let $V = \operatorname{span}\{(1, 0, 1), (0, 1, 0)\}$. Is $(2, 3, 2)$ in V? Is $(1, 1, 0)$ in V? What is $\dim V$?

B. Unit II refresh (Midterm 2 style)

Matrices, Ax, systems, dynamics, products, inverses, rank, determinants.

Definitions and meaning

B1. Give the **column view** formula for Ax when A has columns $a_1, \dots, a_n$.

B2. Translate "$Ax = b$ has a solution" into a statement about $C(A)$.

B3. When does $Ax = b$ have (i) no solution, (ii) a unique solution, (iii) infinitely many solutions? Answer in RREF language.

B4. Write the model for a **linear dynamical system** used in class. How does $A^k x_0$ relate?

B5. Define $C(A)$, $N(A)$, and $\operatorname{rank}(A)$. How do rank and $\dim N(A)$ relate for an $m \times n$ matrix?

B6. For $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, write $\det A$. What does $|\det A|$ mean for the unit square?

B7. True or false: "If $\det A = 0$, then A is invertible." One-line reason.

Computational

B8. Let

$$A = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & 1 \end{bmatrix}, \quad x = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}.$$

Compute Ax using the column-combination view. Input and output dimensions of $x \mapsto Ax$?

B9. Let

$$R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

Compute Re_1 and Re_2. Geometric effect?

B10. Write

$$\begin{aligned} x + 2y &= 5, \\ 3x - y &= 1 \end{aligned}$$

as $Ax = b$ and solve. How many solutions?

B11. The augmented matrix $[A \mid b]$ row-reduces to

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 3 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Identify pivot / free variables and write the general solution.

B12. Let

$$A = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}, \quad x_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Compute $x_1 = Ax_0$ and $x_2 = Ax_1$. Express x_3 using a matrix power.

B13. Let

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Compute AB and BA. Are they equal? Compute A^2.

B14. Find the inverse of $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ if it exists. Is $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ invertible? Two reasons.

B15. The RREF of a 3×4 matrix A is

$$\begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Find $\operatorname{rank}(A)$, a basis for $N(A)$, and check $\operatorname{rank}(A) + \dim N(A) = n$.

B16. Compute $\det \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}$ and $\det \begin{bmatrix} 2 & 1 & 0 \\ 0 & -3 & 4 \\ 0 & 0 & 5 \end{bmatrix}$.

B17. Cost: for large n-vectors a, b, c, compare forming $(ab^{\mathsf{T}})c$ versus $a(b^{\mathsf{T}}c)$.

C. Since Midterm 2 (eigenvectors, complex, least squares, SVD)

Definitions and meaning

C1. What does it mean for nonzero v to be an **eigenvector** of A with eigenvalue λ?

C2. Write the **characteristic equation** for a square matrix A.

C3. Interpret $a + bi$ as a point in the complex plane. Compute $(1 + 2i) + (3 - i)$ and $(1 + 2i)(3 - i)$.

C4. Formulate the **least-squares** problem for $Ax \approx b$. What is the **residual**?

C5. In one sentence: what does a **rank-k SVD approximation** keep, and what does it discard?

C6. True or false, with a one-line reason: "If $Ax = b$ is inconsistent, then least squares has nothing to say."

C7. True or false, with a one-line reason: "Every real 2×2 rotation matrix has a real eigenvector."

Eigenpairs and complex arithmetic

C8. Let

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}, \quad v = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Verify that v is an eigenvector and find λ.

C9. Let

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}.$$

Form $\det(A - \lambda I)$ and find the eigenvalues. For the larger eigenvalue, find one eigenvector.

C10. Let

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

(a) Compute $A \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $A \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Does A look like a rotation? (b) Form $\det(A - \lambda I)$ and show the eigenvalues are not real. (c) One complex eigenvalue is $\lambda = i$. Verify that $v = (1, -i)$ (or an equivalent multiple) satisfies $Av = \lambda v$.

C11. Dynamics connection. Suppose A has eigenpairs (λ_1, v_1) and (λ_2, v_2) with $|\lambda_1| > |\lambda_2|$, and $x_0 = c_1 v_1 + c_2 v_2$ with $c_1 \neq 0$. In one or two sentences: what does $A^k x_0$ look like for large k?

Least squares

C12. Consider the inconsistent system

$$\begin{bmatrix}1\\1\\1\end{bmatrix} x = \begin{bmatrix}1\\2\\2\end{bmatrix}.$$

(Here A is 3×1.)

(a) Why is there no exact solution?
(b) The least-squares $\hat{x}$ minimizes $\|Ax - b\|$. Compute $\hat{x}$ using the normal equation $A^\mathsf{T} A\hat{x} = A^\mathsf{T} b$ (scalars here).
(c) Compute the residual $r = b - A\hat{x}$ and $\|r\|$.

C13. Same A and b as C12. Check that r is orthogonal to the column of A.

C14. Data points $(0, 1)$, $(1, 2)$, $(2, 2)$ are to be fit by a line $y = \beta_0 + \beta_1 t$ in the least-squares sense.

(a) Write the matrix A and vector b in $A \begin{bmatrix}\beta_0\\\beta_1\end{bmatrix} \approx b$.
(b) Form $A^\mathsf{T} A$ and $A^\mathsf{T} b$ (you need not solve unless you want a check).

SVD concepts

C15. Suppose the singular values of a matrix are $20,\ 5,\ 0.4,\ 0.01$.

(a) Roughly how much of the "energy" (squared singular values) is in the top two components?
(b) Why might a rank-2 approximation be attractive for storage?
(c) Name one thing a very low-rank model might miss.

C16. In the factorization $A = U\Sigma V^\mathsf{T}$, which factor carries the singular values? What does "orthogonal factors" buy you conceptually (one sentence)?

C17. Software interpretation. A classmate reports eigenvalues 1.00 and 0.02 for a transition matrix and concludes "after many steps everything goes to zero." What is the more careful reading in terms of dominant eigenbehavior?

D. Mixed cumulative

Show reasoning. These mix units or require choosing a tool.

D1. Which tool? For each task, name a primary tool from: distance / cosine, span / independence / basis, Ax / transform, RREF / free variables, A^k / eigenvalues, least squares / residual, SVD / rank-k, inverse / determinant.

Task	Tool
"Best fit when more equations than unknowns"	
"Compress an image keeping the largest modes"	
"All solutions of an underdetermined system"	
"Will repeated application stretch one direction most?"	
"Nearest catalog item to a query"	
"Does this 2×2 squash area to zero?"	

D2. Let

$$A = \begin{bmatrix}1 & 1\\0 & 1\end{bmatrix}, \quad b = \begin{bmatrix}4\\2\end{bmatrix}.$$

(a) Solve $Ax = b$.

(b) Find eigenvalues of A (hint: triangular).
(c) Interpret A geometrically on the standard basis.

D3. A café stores each day as a 3-vector (coffee, pastry, total) with total = coffee + pastry.

(a) Write A so feature $= Ax$ for $x = (\text{coffee}, \text{pastry})$.
(b) What is $\text{rank}(A)$?
(c) Why can redundant features hurt distance-based methods?

D4. Composition then eigen. Let

$$F = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

(reflection across the y-axis).

(a) Compute $F \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.
(b) Find eigenvalues and eigenvectors of F.
(c) In one sentence: how do the eigen-directions relate to the geometry of reflection?

D5. Overdetermined fit vs exact solve. You have A that is 100×3 with independent columns, and $b \in \mathbb{R}^{100}$.

(a) Why is an exact solution of $Ax = b$ unlikely?
(b) What problem do you solve instead?
(c) What does a small residual tell you? What does a large residual tell you?

D6. Dynamics + least squares flavor. States $x_0 = (1, 0)$, $x_1 = (0.6, 0.4)$, $x_2 = (0.52, 0.48)$ come from $x_{t+1} = Ax_t$ with unknown 2×2 matrix A. Explain in words how you could set up a least-squares problem to estimate the four entries of A from several observed pairs (x_t, x_{t+1}) (no need to solve a large system by hand).

D7. Null space and dependence. For

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix},$$

find nonzero x with $Ax = 0$, write the column dependence, compute $\det A$, and say whether least squares for $Ax \approx b$ has a unique $\hat{x}$ for every $b \in \mathbb{R}^2$.

D8. SVD decision. Two encodings of video frames: (i) store three frames separately and compress each with rank k; (ii) store one frame plus two difference frames and compress those. In two sentences: what quantitative comparisons would you report (error, storage, runtime), and what might make (ii) win?

D9. Hand vs software. For each, say **hand**, **software**, or **either** and why in a few words:

(a) 2×2 characteristic polynomial
(b) eigenvalues of a 50×50 matrix
(c) RREF of a 3×4 augmented matrix from class
(d) rank-5 approximation of a 400×400 image matrix
(e) checking $Av = \lambda v$ for a proposed pair

D10. Cumulative short proofs / reasons (2–4 sentences each).

(a) If v is an eigenvector of A with $\lambda \neq 0$, explain why v cannot be in $N(A)$.
(b) If $\text{rank}(A) < n$ for A that is $m \times n$, explain why $A^{\mathsf{T}}A$ is not positive definite.
(c) Why cosine similarity can prefer a different neighbor than Euclidean distance.

Answers

Check only after a closed-book attempt. Update your definitions/formulas list for every miss.

A — Unit I

A1. $c_1a_1 + \cdots + c_ka_k$ for scalars c_j.

A2. $a^\mathsf{T}b = \sum_i a_ib_i$; orthogonal when the product is 0.

A3. Linear: $f(\alpha x + \beta y) = \alpha f(x) + \beta f(y)$ (equivalently $f(x) = a^\mathsf{T}x$). Affine allows a constant term: $a^\mathsf{T}x + b$.

A4. $\|x\| = \sqrt{x^\mathsf{T}x}$; distance $\|x - y\|$; cosine similarity $a^\mathsf{T}b/(\|a\|\|b\|)$.

A5. Independent spanning list; dimension = number of vectors in any basis.

A6. **False** — vectors can be close in position but point in very different directions (or vice versa for angle vs distance).

A7. (a) **scalar** (b) **set of vectors** (c) **undefined** — scalar plus vector (d) **n-vector**

A8. $u + v = (3, 3, 1)$; $2u - v = (6, -6, 5)$; $u^\mathsf{T}v = -6$.

A9. $w = 4a + 1b$ (since $4(1, 0) + (1, 1) = (5, 1)$).

A10. $\|x\| = \sqrt{6}$; distance $\sqrt{(3)^2 + (-4)^2} = 5$.

A11. $\cos\theta = 1/\sqrt{2}$; unit vector $(3/5, 4/5)$.

A12. **Affine** — nonzero constant term (fails $f(0) = 0$ / superposition).

A13. Distances: to z_1 is 1; to z_2 is 2; to z_3 is 1. Nearest: z_1 or z_3 (tie).

A14. Assign: $x_1, x_2 \to z_1$; $x_3, x_4 \to z_2$. Updated centers: $z_1 = (0, 1)$, $z_2 = (6, 1)$.

A15. **Dependent**: $(-2, -4) = -2(1, 2)$.

A16. $(2, 3, 2) = 2(1, 0, 1) + 3(0, 1, 0)$ **yes**; $(1, 1, 0)$ **no** (any vector in V has equal first and third entries). $\dim V = 2$.

B — Unit II

B1. $Ax = x_1a_1 + \cdots + x_na_n$.

B2. $b \in C(A)$.

B3. (i) inconsistent row $[0 \cdots 0 \mid c]$ with $c \neq 0$; (ii) consistent, no free variables; (iii) consistent with a free variable.

B4. $x_{t+1} = Ax_t$; after k steps, $x_k = A^kx_0$.

B5. $C(A) = \text{span}\{\text{columns}\}$; $N(A) = \{x : Ax = 0\}$; rank $= \dim C(A) =$ number of pivots; $\dim N(A) = n - \text{rank}(A)$.

B6. $ad - bc$; $|\det A|$ is the area scaling of the unit square.

B7. **False** — $\det A = 0$ means noninvertible.

B8. $Ax = (-1, 2)$; input $\mathbb{R}^3$, output $\mathbb{R}^2$.

B9. $Re_1 = (0, 1)$, $Re_2 = (-1, 0)$; 90° counterclockwise rotation.

B10. $A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$, $b = (5, 1)$; unique solution $x = 1$, $y = 2$.

B11. Pivots x_1, x_2; free $x_3 = t$; $x = (3 - 2t,\ 1 + t,\ t)$.

B12. $x_1 = (0.5, 0.5)$; $x_2 = (0.5, 0.5)$; $x_3 = A^3 x_0$.

B13. $AB = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}$, $BA = \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix}$; not equal; $A^2 = \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}$.

B14. Inverse $\begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$. Second matrix **not** invertible: $\det = 0$; columns dependent / nontrivial null space.

B15. Rank 2; free $x_2 = s$, $x_4 = t$;

$$x = s \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3 \\ 0 \\ 1 \\ 1 \end{bmatrix};$$

$\dim N(A) = 2$; $2 + 2 = 4$.

B16. 10; -30.

B17. Outer-product-first builds an $n \times n$ matrix; $a(b^\mathsf{T} c)$ only needs a dot product then a scale — much cheaper.

C — Since Midterm 2

C1. $Av = \lambda v$ with $v \neq 0$.

C2. $\det(A - \lambda I) = 0$.

C3. Point (a, b) in the plane; sum $4 + i$; product $(1)(3) + (1)(-i) + (2i)(3) + (2i)(-i) = 3 - i + 6i - 2i^2 = 5 + 5i$.

C4. Minimize $\|Ax - b\|$ over x; residual $r = b - A\hat{x}$ (or $Ax - b$ with sign convention stated).

C5. Keeps the k largest singular components; discards the rest (smaller modes / noise / fine detail).

C6. False — least squares is exactly for inconsistent / overdetermined cases.

C7. False — 90° rotation has no real eigenvector (eigenvalues $\pm i$).

C8. $Av = (0, 3) = 3v$, so $\lambda = 3$.

C9. $A - \lambda I = \begin{bmatrix} 4 - \lambda & 1 \\ 2 & 3 - \lambda \end{bmatrix}$; $\det = (4 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 7\lambda + 10 = 0$; $\lambda = 5, 2$. For $\lambda = 5$: $(A - 5I)v = 0$ gives $\begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} v = 0$, so $v = (1, 1)$ (or multiple).

C10. (a) $(0, 1)$ and $(-1, 0)$; yes, 90° CCW rotation. (b) $\det(A - \lambda I) = \lambda^2 + 1 = 0$, so $\lambda = \pm i$. (c) $Av = (i, 1)$ and $iv = (i, -i^2) = (i, 1)$; match.

C11. $A^k x_0 = c_1 \lambda_1^k v_1 + c_2 \lambda_2^k v_2$ is dominated by the $\lambda_1^k v_1$ term (direction of v_1, scaled by λ_1^k).

C12. (a) b not a multiple of $(1, 1, 1)$ (entries not equal). (b) $A^\mathsf{T} A = 3$, $A^\mathsf{T} b = 5$, so $\hat{x} = 5/3$. (c) $A\hat{x} = (5/3, 5/3, 5/3)$; $r = (-2/3, 1/3, 1/3)$; $\|r\| = \sqrt{2/3} = \sqrt{6}/3$.

C13. Column is $(1, 1, 1)$; $r^\mathsf{T}(1, 1, 1) = -2/3 + 1/3 + 1/3 = 0$.

C14. (a)

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

(b) $A^\mathsf{T} A = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}$, $A^\mathsf{T} b = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$.

C15. (a) Energy $\propto 20^2 + 5^2 + 0.4^2 + 0.01^2 = 400 + 25 + 0.16 + 0.0001$; top two are $425/425.16 \approx 99.96\%$. (b) Store two singular triples / factors instead of the full matrix. (c) Fine detail, small features, or structured residuals in the discarded modes.

C16. Σ (diagonal of singular values). Orthogonal U, V give orthonormal bases for row/column spaces / preserve norms in those coordinates.

C17. The mode with $|\lambda| \approx 1$ dominates; the state tends toward that eigen-direction (not necessarily to zero). The 0.02 mode dies out.

D — Mixed

D1. Least squares / residual; SVD / rank-k; RREF / free variables; A^k / eigenvalues; distance / cosine; inverse / determinant.

D2. (a) $x_2 = 2$, $x_1 = 2$. (b) Both eigenvalues 1 (diagonal of triangular matrix). (c) $e_1 \mapsto e_1$, $e_2 \mapsto (1, 1)$ — shear.

D3. (a) $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$. (b) Rank 2. (c) Redundant coordinate double-counts / distorts distances.

D4. (a) $(-1, 2)$. (b) Eigenvalues ± 1: e.g. $\lambda = 1$ with $(0, 1)$ (fixed y-axis); $\lambda = -1$ with $(1, 0)$ (reversed x-axis). (c) Eigen-directions are the axis that stays put and the axis that flips.

D5. (a) b almost surely not in the 3-dimensional column space. (b) $\min_x \|Ax - b\|$. (c) Small residual: data nearly consistent with the model; large residual: model misspecification or large noise / wrong features.

D6. Each pair gives $x_{t+1} \approx Ax_t$, i.e. two linear equations in the four unknown entries of A. Stack many timesteps into one big overdetermined system $Ma \approx y$ for the vectorized entries a of A, then solve in the least-squares sense.

D7. e.g. $x = (2, -1)$; $2c_1 - c_2 = 0$; $\det A = 0$; least-squares $\hat{x}$ is **not** unique for every b (columns dependent $\Rightarrow A^\mathsf{T}A$ singular; infinitely many minimizers when any exist).

D8. Report reconstruction error (e.g. RMS / Frobenius), storage for the kept factors, and factorization / reconstruction time. Differences are often more compressible (singular values decay faster), so (ii) can win on storage/error for the same k.

D9. (a) hand (or either); (b) software; (c) hand / either; (d) software; (e) either (hand for tiny; software numerically).

D10. (a) $Av = \lambda v \neq 0$, so $v \notin N(A)$. (b) Independent columns fail $\Rightarrow$ nonzero x with $Ax = 0 \Rightarrow x^\mathsf{T}A^\mathsf{T}Ax = 0$ with $x \neq 0$, so not positive definite. (c) Distance cares about position; cosine cares about direction — a long vector nearly parallel to the query can beat a closer but angled neighbor under cosine, and vice versa under distance.